Palaeomagnetic investigations of the Tertiary and Quaternary igneous rocks: V. The basic Tertiary basalts of Lower Silesia, Poland *)

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With 7 figures and 1 table

Zusammenfassung

An Proben aus vulkanischen Eruptivgesteinen von 69 Lokalitäten in Niederschlesien nahmen die Autoren Messungen der Richtungen remanenter Magnetisierung vor. Die Vulkantätigkeit umfaßte die Zeit vom Oligozän bis Pliozän oder Pleistozän. Die magnetischen Daten lassen vermuten, daß der Vulkanismus in Polen und Ostdeutschland gleichzeitig einsetzte. Die Stratigraphie ist noch zu wenig bekannt, um eine Folge magnetischer Zonen aufzustellen.

Abstract

Measurements of the directions of remanent magnetization of samples from sixty nine sites in the volcanic igneous rocks of Lower Silesia are reported. The activity spanned the time from Oligocene to Pliocene or Pleistocene, the magnetic data suggest the possible equivalence of the onset of vulcanity in Poland and East Germany. The stratigraphic control is too poor to permit the distinction of a sequence of magnetic zones.

Résumé

Les mesures de l'aimantation rémanente sur les échantillons prélevés de 69 sites dans les roches volcanique de Basse Silésie ont été faites. L'activité volcanique a duré de l'Oligocène jusqu'au Pliocène ou même Pleistocène; les résultats magnétiques indiquent qu'il est possible que le commencement du volcanisme en Pologne et Allemagne de l'est date de la même époque. La stratigraphie est mal connue et il n'est pas encore possible d'établir une série de zones magnétiques.

Краткое содержание

Исследовали направления остаточного магнетизма вулканических эруптивных пород в нижней Силезии. Вулканическая деятельность здесь продолжалась от олигоцена до плиоцена или плейстоцена. На основании этих палеомагнетических измерений можно утверждать, что в Польше и Восточной Германии вулканическая деятельность началась одновременно. Для подразделения на палеомагнитные зоны имеющиеся стратиграфические данные недостаточны.

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Introduction

The Tertiary basalts of Lower Silesia form the eastern extremity of the North Bohemian province. Igneous rocks belonging to this province stretch from Doupovské Hory (Duppauer Gebirge) in the west through the region of Teplice, Ustí and Dečín in northern Bohemia and the Lausitz area of Germany into Poland (fig. 1). To the southwest this belt of alkali volcanics is represented by outcrops which extend as far as the Rhine valley, e.g. the Kaiserstuhl and Hegau. In Poland outcrops are most abundant in the vicinity of Lubań and Zgorzelec, between Lwówek Śląski and Jawor and between Gryfów Śląski and Świeradów. More scattered outcrops are found around Strzegom, Niemcza, Strzelin and Lądek, with isolated occurrences between Strzelin, Strzelce Opolskie and Głubczyce. The total number of outcrops of volcanic rocks, basalts and basaltic tuffs, found in Lower Silesia is estimated at between 200 and 300, of which less than one fifth are lava flows. Most of the outcrops are to be found on 1 : 25,000 geological maps published prior to World War II. Although short descriptions were presented in the explanatory texts detailed observations were seldom published (e.g. Möhl 1875, Krusch 1895, Berg 1930, Berger 1932, Ebert 1937, 1939). Of more recent date is the synthesis of field and laboratory data of 135 exposures of basalts by Wojno, Pentlakowa & Szarras (1951). Smulikowski (1960) classified the Silesian volcanic rocks into basalts, basanites, tephrites, nephelinites, limburgites, basanitoids and tephritoids with additional local varieties. Other recent petrological investigations include those of Jerzmański (1956, 1961, 1965), Pendias & Maciejewski (1959), Kozłowski & Parachoniak (1960) and Chodyniecka (1967).

Between the summer of 1962 and the spring of 1965 oriented samples from sixty nine sites in the basalts of Lower Silesia were collected for palaeomagnetic work. The sites were widely distributed (see fig. 2). The palaeomagnetic results of this investigation are reported here and represent an extension of the work previously reported (Nairn & Vollstädt 1967). The detailed site geological descriptions will appear subsequently in a joint paper with Jerzmański in the Annales de la Société Géologique de Pologne.

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Outline of geology

a) Age relationships.

In common with the rest of the north Bohemian province and with the rest of western Europe, the basalt activity in Lower Silesia was closely related to young crustal movements. In western Poland these movements principally involved vertical block movements of the Sudeten Mountains. Basaltic rocks occur on both the uplifted Sudetic horst and on the down-

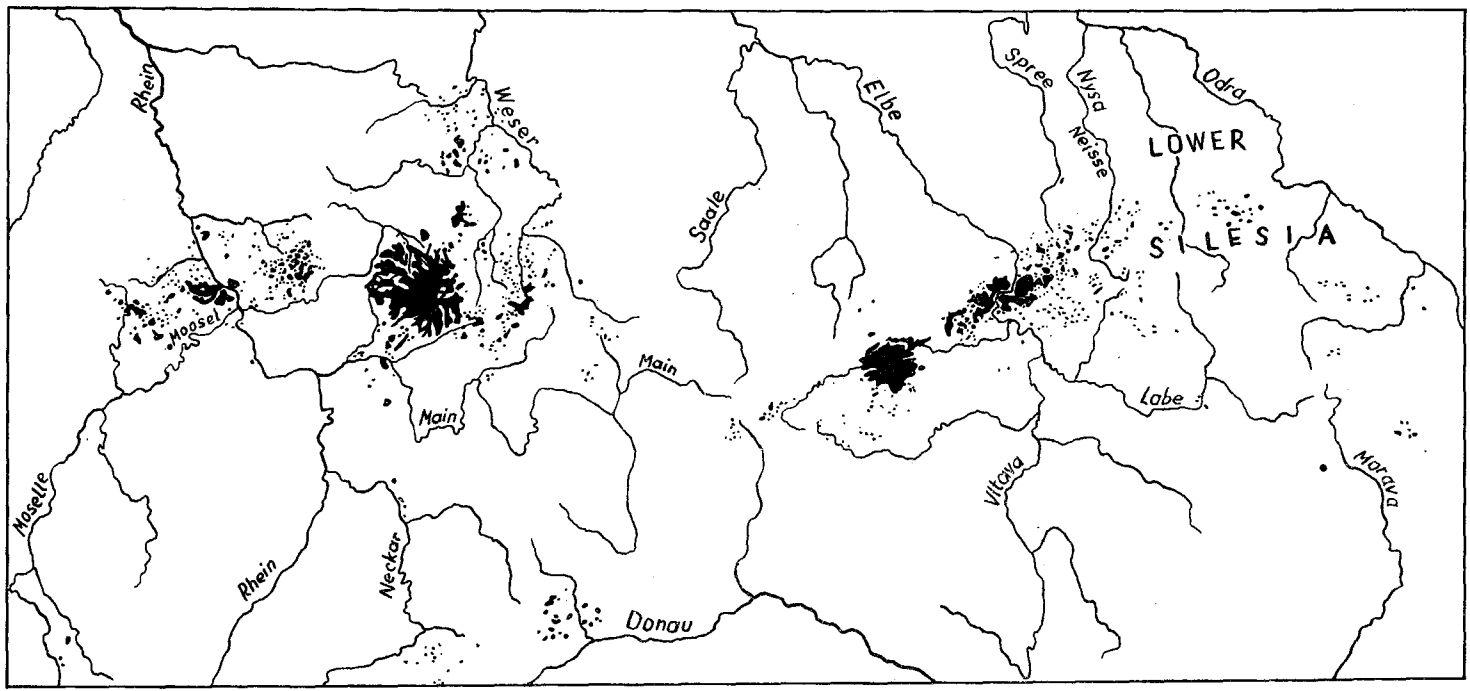


Fig. 1. Position of the Lower Silesian basalts within the Central European Tertiary volcanic province (after KOPECKÝ 1966 with additions by the authors).

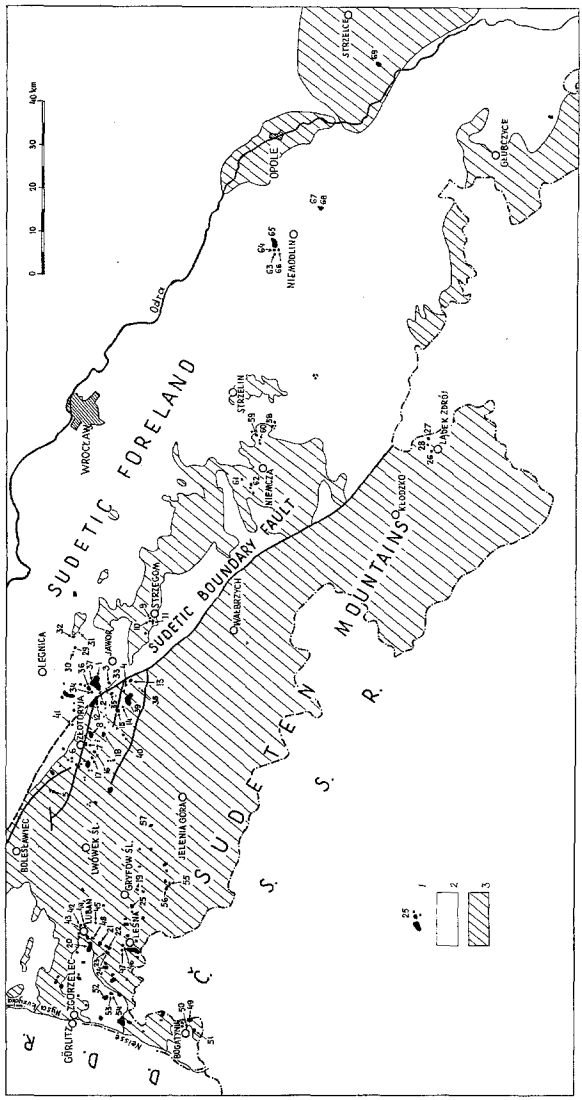


Fig. 2. Sampling sites in the Lower Silesian Tertiary basalts. 1. Basalt sites with locality numbers corresponding to those in table, 2. Neogene sediments, 3. pre-neogene rocks.

thrown Sudetic foreland. As a result of prolonged erosion during Tertiary and Quaternary times only relicts of formerly more extensive volcanic material mostly in the form of plugs, vents and dykes remain on the uplifted horst. Lava flows such as found near Złotoryja and Lądek are comparatively rare. As might be expected, preservation is better in the downthrown area northeast of the Sudetic boundary fault, where in addition to lava flows, volcanic cones and calderas can be recognized.

In the absence of a detail programme of radiometric dating, and the general paucity of stratigraphic data, it is at present very difficult, if possible at all, to reconstruct the succession of volcanic events for the whole Lower Silesian sub-province. There are nevertheless a number of control points which enable the time range of the vulcanicity to be defined.

In the vicinity of Meçinka, west of Jawor, the trachyandesite flow overlies Upper Eocene sediments, and is covered by Upper Oligocene according to palaeobotanic work (Jerzmański 1961). This is consistant with the He age determinations of Urry (1936) which gave values of 29 ± 2 and 34 ± 2 m.y. for the flow. South of Meçinka samples of the Bazaltowa columnar basalt gave Urry 36 ± 2 m.y.

Basalt lava flows are known within Miocene rocks in which lignite occurs. They are found in the region of Lubań, Zgorzelec, Bogatynia and Gryfów Śląski. Between flows are red clays the result of contemporary weathering of the flows (cf. Kozłowski & Parachoniak 1960, Birkenmajer 1967). These Miocene rocks are cut by basaltic sills found near Leśna and Gryfów Śląski. The youngest activity is represented in this area by volcanic vents filled with breccia which cut through all of the preceeding. Fragments of red clay and lava are found in the vent breccias.

Near Lądek a basanite lava rests on the Pliocene fluvial gravels of the Biała Lądecka river (cf. Walczak 1957), suggesting a Pliocene or Pleistocene age for the lava (Berger 1932, Smulikowski 1960).

The volcanic activity therefore began in Oligocene times, at about 36 ± 2 m.y., reached a maximum during the Miocene, finally dying out either in late Pliocene or Pleistocene. It is not possible to say at this stage whether the maximum of volcanic activity occurred at the same time as the maximum in northern Bohemia (Lower Aquitanian to Burdigalian, Kopecký 1966) or whether it was somewhat later.

b) Volcanic forms.

The lava flows are generally in the form of flat lying basaltic sheets dissected by erosion forming gentle hills and plateaux. They occur especially between Lubań and Leśna, between Złotoryja, Legnickie Pole and Jawor, then east of Niemcza, south of Niemodlin and near Lądek. The hill slopes are often softer sedimentary beds of the lignite bearing formation in which pyroclastic layers are interbedded. The lava sheets may exceed 100 m in thickness although the original scoriaceous surface of the sheets is seldom preserved. Columnar jointing is common with individual columns often over a dozen metres in length and with column thicknesses of 0.5—1.5 m. Although less regular than those developed in plugs and vents they can be observed to be vertical or nearly vertical, normal to the cooling surface.

More common than flows are the plugs and vents. In some cases they may be associated with lava flows, and be recognized as different units within a single strato volcano. They may be round, elliptical or irregular in cross-section. Some are located along dislocations parallel or transverse to the Sudetic boundary fault between Jawor and Złotoryja. They are

Table 1

Site No.	Location and Lithology	Magnetization D	I	Sampling a	b	Cofidence a
1	Meçinka W trachyandesite lava 29 ± 2 my	154.7	— 56.6	4	— 4	8.1
2	Meçinka E trachyandesite lava 34 ± 2 my	171.2	— 53.9	4	— 5	4.3
3	Winnica trachyandesite plug	177.2	— 53.4	6	— 6	3.3
4	Rataj basalt plug	334.5	+ 70.3	6	— 6	11.1
5	Grodziec nephelinite cone	12.3	+ 30.5	4	— 5	5.5
6	Pielgrzymka basalt plug	177.2	— 77.8	6	— 6	10.0
7	Wilcza Góra basanite plug	131.7	— 85.9	6	— 6	7.1
8	Kozia Góra plagioclase basalt plug with glass	351.9	— 26.7	5	— 5	6.8
9	Strzegom-Kreuzberg I basalt plug	285.8	— 83.8	6	— 6	4.3
10	Strzegom-Georgenberg II basalt plug	285.8	— 85.6	6	— 6	5.2
11	Strzegom-Breitberg III basalt plug	142.1	— 84.1	6	— 6	8.2
12	Sichów basanite plug	248.5	+ 75.8	6	— 6	6.2
13	Bazaltowa trachybasanite plug 36 ± 2 my	159.7	— 63.7	6	— 6	7.1
14	Owcza Góra basalt plug	187.6	— 59.8	6	— 6	4.5
15	Czartowska Skala nephelinite	17.6	+ 73,3	6	— 6	4.2
16	Krzeniów I basalt plug	12.3	+ 71.4	5	— 6	5.3
17	Krzeniów II plagioclase basalt with glass plug (basanitoid?)	132.7	— 51.8	6	— 6	32.3
18	Krzeniów III nephelinite plug	no confidence		—		—
19	Leopoldówka limburgite plug	14.4	+ 72.2	6	— 6	7.3
20	Luban I nephelinite lava flow	195.8	+ 20.1	4	— 6	25.7
21	Luban II nephelinite lava flow	no confidence		—		—
22	Luban III nephelinite plug	102.5	+ 53.1	6	— 6	12.2
23	Zareba Górna I upper flow nephelinite	176.4	— 51.3	6	— 6	9.9
24	Zareba Górna II nephelinite lower lava flow	159.8	— 46.3	6	— 6	17.7
25	Wieza complex sill intrusion basanite	348.3	+ 65.7	6	— 8	5.1
26	Ladek Zdrój basanite lava	3.6	+ 66.6	7	— 7	4.3
27	Lutynia I pyroxene basalt with glass	352.8	+ 65.3	6	— 7	4.0
28	Lutynia II basanite	356.1	+ 65.4	7	— 7	4.2
29	Lubien I—II plagioclase basalt lava (trachyandesite?)	25.9	+ 60.5	8	— 8	3.1
30	Pawlowice plagioclase basalt plug with glass (basanitoid?)	333 3	+ 71.9	6	— 6	16.7
31	Mikolajowice I nephelinite lava flow	322.4	+ 81.0	5	— 6	6.1
32	Mikolajowice II basalt plug	202.0	— 47.0	7	— 7	11.7
33	Mecinka S. basalt plug	358.1	+ 13.9	6	— 6	14.4
34	Chróslica N trachyandesite lava	no result		0	— 5	—
35	Górzec trachyandesite plug	106.0	— 43.4	6	— 6	no confi.
36	Slup trachyandesite lava	129.1	— 86.6	6	— 6	9.6
37	Zarek trachyandesite lava	161.7	— 50.9	6	— 6	9.1
38	Myslinów basalt plug	179.1	— 64.2	5	— 6	12.2
39	Muchów trachybasanite plug	189.6	— 54.4	6	— 7	3.2
40	Rzeszówek nephelinite plug	146.5	— 46.7	5	— 6	no confi.
41	Kozów nephelinite lava	346.1	+ 75.8	6	— 6	6.3
42	Uniegoszcz I nephelinite lava	164.4	+ 26.6	6	— 6	no confi.
43	Uniegoszcz II nephelinite plug	169.3	— 15.4	5	— 6	no confi.
44	Uniegoszcz III nephelinite plug	176.3	— 74.1	6	— 6	9.1
45	Jalowiec nephelinite plug	170.0	— 73.1	6	— 6	10.0
46	Lesna I basalt lava	161.4	— 56.5	6	— 6	4.2
47	Lesna II basalt plug	352.4	+ 52.5	5	— 6	9 4
48	Ksieginki nephelinite lava	no result		0	— 7	—

C.S.D.	C.S.E.	Vector	Demag.	Log Mean Intensity & St. dev × 10^{-4} cm				Log Mean Suscept × 10^{-4}cm	
K*		R	field	NRM		demag.		×	S.D.
129.1		3.977	170	1.77	0.74	1.04	0.47	1.42	0.84
465.4		3.994	255	1.41	0.07	1.24	0.77	1.38	0.65
402.1		5.988	85	1.07	0.77	1.11	0.66	1.49	0.59
37.4		5.867	85	1.74	1.36	1.57	1.25	1.29	0.96
279.0		3.989	170	1.12	0 46	0.90	0.36	1.47	0.88
45.5		5.890	85	1.87	1.41	1.61	1.33	1.05	0.77
89.0		5.944	85	1.19	0.77	1.07	0.36	1.44	1.14
129.1		4.969	85	1.30	0.67	1.18	0.73	1.32	0.70
247.0		5.980	255						
169.7		5.971	170						
68.2		5.927	255						
116.3		5.957	85						
90.0		5.945	170						
224.8		5.978	85						
261.6		5.981	85						
208.2		4.981	85						
5.3		5.050	340						
—		—	—	—		—		—	
85.7		5.942	85						
13.8		3.783	85						
—		—	—	—		—		—	
31.1		5.839	340						
46.8		5.893	85						
15.3		5.673	85						
177.0		5.972	0						
5.8	2.2	6.969	85	1.52	0.15	1.37	0.08	1.19	0.13
4.9	2.0	5.982	0	1.83	0.34	1.83	0.34	1.36	0.09
5.7	2.2	6.970	85	1.59	0.15	1.41	0.19	1.70	0.38
4.6	1.6	7.978	170	1.50	0.05	1.39	0.08	1.45	0.47
19.7	8.1	5.706	85	1.22	0.11	0.99	0.15	1.54	0.29
6.4	2.9	4.975	85	1.33	0.18	1.10	0.16	1.76	0.46
15.4	5.8	6.783	85	1.09	0.12	0.84	0.14	1.67	0.23
17.1	7.0	5.778	0	1.25	0.13	1.25	0.13	1.48	0.22
—	—	—	—	1.04	0.12	—	—	1.52	0.25
61.2	25.0	3.370	170	1.95	0.50	1.10	0.16	1.89	0.34
11.5	4.7	5.899	85	0.79	0.15	0.76	0.16	1.72	0.39
10.9	4.4	5.910	170	1.34	0.13	1.02	0.10	1.56	0.12
12.8	5.7	4.901	170	1.89	0.40	1.46	0.30	1.86	0.45
3.9	1.6	5.989	85	1.39	0.11	1.23	0.05	1.42	0.02
40.9	18.3	4.014	85	1.18	0.16	1.27	0.08	1.73	0.03
7.6	3.1	5.957	170	1.42	0.27	0.83	0.12	1.83	0.02
44.9	18.3	4.527	170	1.33	0.42	0.64	0.08	1.70	0.05
34.7	15.5	4.283	170	1.32	0.22	0.89	0.22	1.66	0.12
10.9	4.4	5.910	85	1.43	0.15	1.42	0.13	1.96	0.14
12.0	4.9	5.891	85	1.24	0.21	1.33	0.05	2.11	0.01
5.1	2.1	5.980	85	1.60	0.10	1.45	0.13	1.63	0.51
9.9	4.4	4.940	255	1.65	0.11	1.21	0.11	1.40	0.22
—		—	—	0.77	0.16	—	—	1.87	0.08

Table 1 (Continuation)

Site No.	Location and Lithology	Magnetization D	I	Sampling a	b	Confidence a
49	Markocice lava?	78.1	+ 48.9	5	— 6	no conf.
50	Bogatynia hornblende basanite lava?	18.3	+ 63.1	5	— 6	17.3
51	Opolno Zdrój trachyphonolite plug	315.0	+ 37.6	8	— 8	no conf.
52	Sulików nephelinite plug	167.8	— 69.2	7	— 7	3.0
53	Mala Wies Górna I nephelinite plug	208.3	— 56.2	6	— 6	2.6
54	Mala Wies Górna II plagioclase basalt plug with glass	6.5	+ 72.0	6	— 6	9.4
55	Proszowa I nephelinite plug	236.6	— 76.4	6	— 6	9.6
56	Proszowa II nephelinite plug	222.3	— 63.7	5	— 6	14.6
57	Pilchowice nephelinite plug	306.6	+ 71.1	3	— 7	12.6
58	Targowica "plagioclase basalt" lava	196.1	— 60.4	6	— 6	8.2
59	Kowalskie-Zelowice I nephelinite lava	215.8	— 44.5	6	— 6	8.6
60	Kowalskie-Zelowice II nephelinite plug	218.3	— 53.6	6	— 6	7.0
61	Gilów I basalt	150.4	— 67.9	4	— 6	17.6
62	Gilów II basalt vent or lower part of cone	184.7	— 72.0	6	— 6	11.0
63	Radoszowice nephelinite plug	45.9	+ 49.2	6	— 6	no conf.
64	Grace I basanite plug	47.9	+ 53.0	3	— 6	11.1
65	Gracze II basanite vent fill of strato-cone	31.6	+ 54.7	6	— 6	4.9
66	Rutki basalt plug	25.8	+ 49.2	6	— 6	6.3
67	Ligota Tulowicka I basanite lava	179.3	— 44.7	5	— 6	7.3
68	Ligota Tulowicka II nephelinite lava	173.6	— 44.0	6	— 6	6.3
69	Góra Sw. Anny nephelinite vent fill of crater or caldera	213.2	— 54.1	6	— 6	5.5

usually characterized by a system of well developed regular columnar jointing mostly vertical in the upper part but spreading fan-wise in the lower part towards laterally distributed cooling surfaces. Feeder veins or dykes cutting the plugs have horizontally disposed joints. In the case of multiple events different columnar patterns may develop within a single vent or plug.

Volcanic cones as such are nowhere found, but it is possible to discern relicts of them, often with a fill of lava or of tuff breccias and agglomerates. In some stratified cones lava flows alternating with tuff and breccia horizons are found e.g. near Gracze (Birkenmajer 1967). In addition dykes and sills showing no direct relationship to any of the volcanic forms described also occur. Where the sills intrude almost unconsolidated sediments as in the Zittau basin intense synintrusive tectonics are observed.

The petrological terminology (table 1) adopted in this paper follows that of Smulikowski and, in part, that of Wojno and Pentalakowa. Field descriptions of the sites have been given by Jerzmański (1956, 1961, 1965), Kozłowski & Parachoniak (1960), Śliwa (1967) and Birkenmajer (1967) and will be amplified in Birkenmajer, Jerzmański & Nairn (in preparation).

C.S.D. K	C.S.E.	Vector R	Demag. field	Log Mean Intensity & St. dev $\times 10^{-4}$ cm NRM		demag.		Log Mean Suscept $\times 10^{-4}$ cm $\times$	S.D.
40.5	18.1	4.034	85	1.09	0.21	0.67	0.10	1.92	0.04
17.9	8.0	4.805	85	1.60	0.12	1.12	0.06	1.96	0.04
47.3	16.7	5.732	85	0.34	0.13	0.25	0.03	1.82	0.03
4.0	1.5	6.985	85	1.27	0.07	1.35	0.07	2.09	0.03
3.2	1.3	5.992	85	1.61	0.06	1.60	0.03	1.66	0.12
11.2	4.6	5.904	85	1.41	0.12	1.35	0.10	1.71	0.37
11.6	4.7	5.899	85	1.37	0.09	0.98	0.19	2.11	0.01
15.2	6.8	4.859	85	1.26	0.12	1.11	0.11	2.08	0.02
8.2	4.7	2.979	255	1.14	0.07	0.94	0.06	1.72	0.56
9.8	4.0	5.927	255	1.62	0.24	1.44	0.31	1.33	0.10
10.4	4.2	5.912	85	1.35	0.32	1.30	0.22	1.43	0.30
8.5	3.5	5.946	0	1.10	0.05	1.10	0.05	1.29	0.12
15.3	7.6	3.894	255	2.18	0.49	0.83	0.14	1.45	0.37
13.1	5.4	5.869	0	1.24	0.17	1.24	0.17	1.51	0.42
25.8	10.5	5.580	85	0.47	0.14	0.42	0.13	1.62	0.48
7.3	4.2	2.984	170	0.92	0 08	0.40	0.12	1.76	0.33
6.0	2.4	5.973	85	1.34	0.21	1.22	0.22	1.79	0.04
7.6	3.1	5.956	85	1.10	0.07	0.71	0.13	1.71	0.05
7.7	3.5	4.964	85	1.24	0.30	1.26	0.28	1.93	0.04
7.6	3.1	5.957	0	1.22	0.14	1.22	0.14	1.69	0.20
6.6	2.7	5.967	0	1.50	0.14	1.50	0.14	1.02	0.17

Palaeomagnetic investigation

a) Field and preparatory techniques.

From each of the sixty nine sites, six or more oriented blocks were collected. Whenever possible advantage was taken of quarries for sampling to ensure fresh material. Sampling was spaced over anything from 20 m to more than 100 m depending upon the nature of the outcrop. From each sample a single core was drilled with non magnetic tools and this core used for palaeomagnetic work.

b) Measurements of remanence directions.

The measurement of the direction of natural remanent magnetization was carried out under a short period astatic magnetometer. All samples were subsequently demagnetized in alternating fields of up to 340 oe. Higher fields, up to about 700 oe, were occassionally used to obtain more complete data. The mean site direction of magnetization was computed after each stage. The direction adopted as the best representation of the field believed to have existed at the site during emplacement was that at which the circle of confidence was a minimum (maximum K or minimum circular standard error). The limitation imposed was that where the minimum scatter was the natural remanent magnetization, subsequent demagnetization should not significantly alter the mean value. The mean direc-

tions are given with other relevent statistics in table 1; and illustrated in fig. 3.

c) Intensity of magnetization.

The intensity of magnetization was measured concurrently with the measurement of the direction of remanence. The site mean intensity of

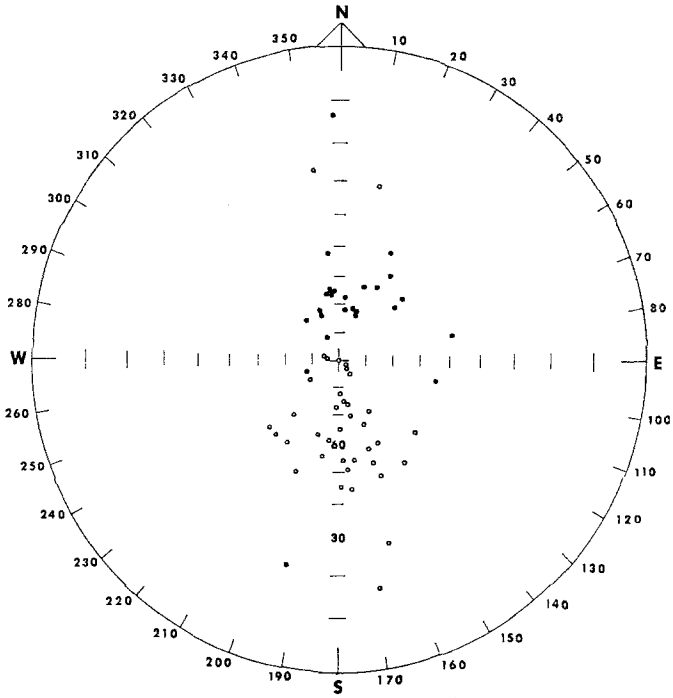


Fig. 3. Site mean directions of magnetization of Lower Silesian basalts. Following standard convention open circles represent projections on the upper hemisphere (reversed magnetization), solid points on the lower hemisphere (normal magnetization).

N. R. M. and at the stage of minimum dispersion where computed are listed in table 1. As the distribution of intensity is not arithmetic (see fig. 6), the mean intensities and standard deviations are given in logarithmic form. Sample of normalized demagnetization curves are figured; fig. 4 shows the results of alternating field demagnetization and fig. 5 the results of thermal demagnetization.

The distribution of intensity of magnetization in the suite of rocks is shown in fig. 6. Essentially this curve shows that the magnitude of secondary magnetization seems to bear a definite relationship to the N. R. M. intensity.

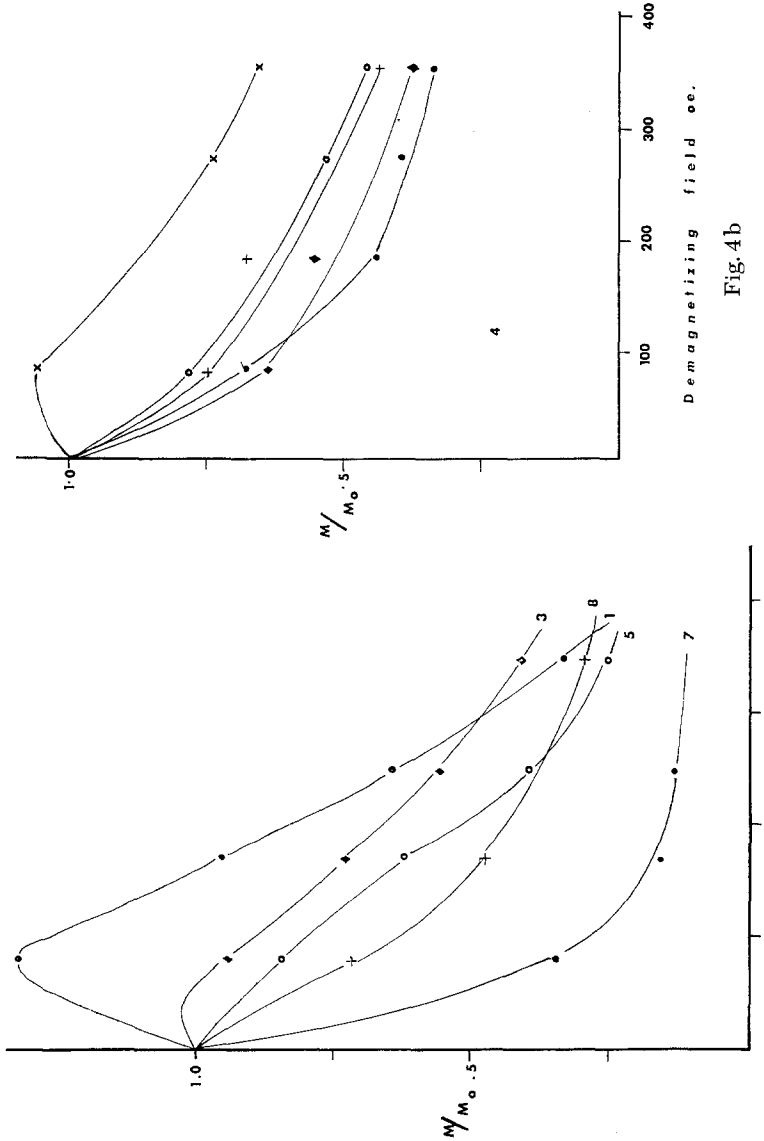


Fig. 4. Normalized alternating field demagnetization curves of Lower Silesian basalt samples. a) Single samples from each of five sites. b) Five samples from one site. The numbers refer to the sites a listed in table 1.

d) Susceptibility of magnetization.

Volume susceptibility was measured on a transformer bridge designed by COLLINSON et al. (1963). Site mean values and standard deviations where completed are given in logarithmic form in table 1, the distribution of measurements is shown in fig. 7.

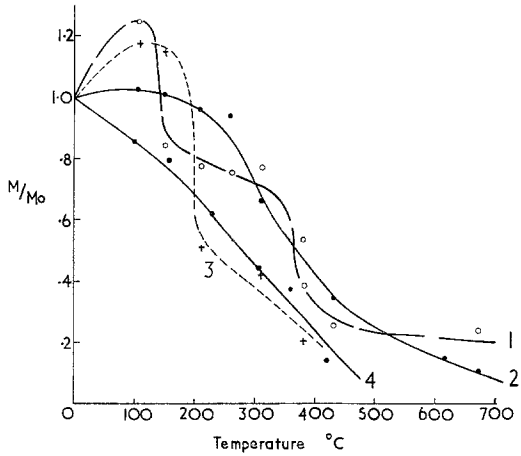


Fig. 5 a. Normalized thermal demagnetization curves from three Lower Silesian sites. The numbers correspond to the sites as listed in table 1.

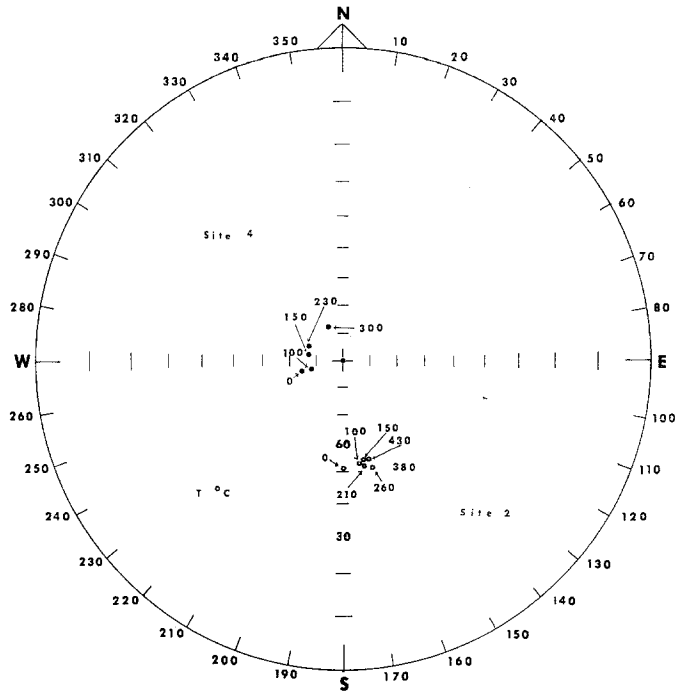


Fig. 5 b. The changes in direction of magnetization of samples from two sites during the course of thermal demagnetization. Normalized intensity curves are illustrated in fig. 5 a.

Discussion of results

a) Directions of magnetization.

The reversely magnetized sites appear to outnumber the normally magnetized sites about 2 : 1. There are also a number of sites with anomalous magnetization, either very low inclination values (e. g. 33) or anomalous sign of inclination with respect to declination (e. g. 12, 22) or both (e. g. 20, 42). The general interpretation of the sequence of magnetic signs is impossible not only because of the general absence of stratigraphic controls and radiometric dates but because of the impossibility of separating magnetic events from epochs. Within this framework of uncertainty there are few clear indications of stratigraphic utility.

1. The flow of Lądek Zdrój (26) is said to be upper Pliocene or Pleistocene. As the flow is normally magnetized, provided it was not extruded during the two short events in the Matuyama reversed epoch (see COX, DOELL, DALRYMPLE 1964), then the flow must be either younger than 0.69 m. y. or older than 2.5 m. y. The same conclusion applies to sites 27, 28. This is a useful restriction of the ages possible.

2. The oldest rocks, both stratigraphically and radiometrically at Meçinka (1, 2) and Bazaltowa (13) are all reversely magnetized. This is the same sign as found by NAIRN & VOLLSTÄDT (1967) in basalts of the Lausitz area which were, on stratigraphical evidence the oldest of the volcanic sequence measured, and of an Oligocene-Miocene age. In an earlier examination of rocks from the Westerwald (NAIRN 1962) it was shown that a change in sign from reversed to normal occurred near the top of middle Chattian. It is therefore tempting to assume all these rocks as pre upper middle Chattian. If, however, the He dates are in error, the rocks could have been extruded much earlier in the Oligocene without contradicting the stratigraphical evidence.

3. The lavas whose association with Miocene (unspecified) lignitic series gives an approximate age, and the later intrusives and vents add little to our knowledge of the Miocene. They indicate a minimum of two changes of sign of magnetization within the Miocene, however NAIRN (1967) has already indicated the possibility of three changes within the upper Miocene.

b) Intensity and susceptibility.

The parallelism of the distribution of the N. R. M. and demagnetized intensity of magnetized curves is of interest in that it suggests that there is a relationship, within a given suite of rocks, between the magnitude of the N. R. M. and the secondary magnetization. The demagnetized intensity distribution is composed of results obtained from rocks to which demagnetizing fields ranging from 0 to 340 oe may have been applied. It is also clear that within one site there may be a considerable variation in intensity and the demagnetization curves may be quite distinct (fig. 4 b). The cause appears to be less the variation in total ferromagnetic content than of position, as is clear from the work of ANGENHEISTER & TURKOWSKY

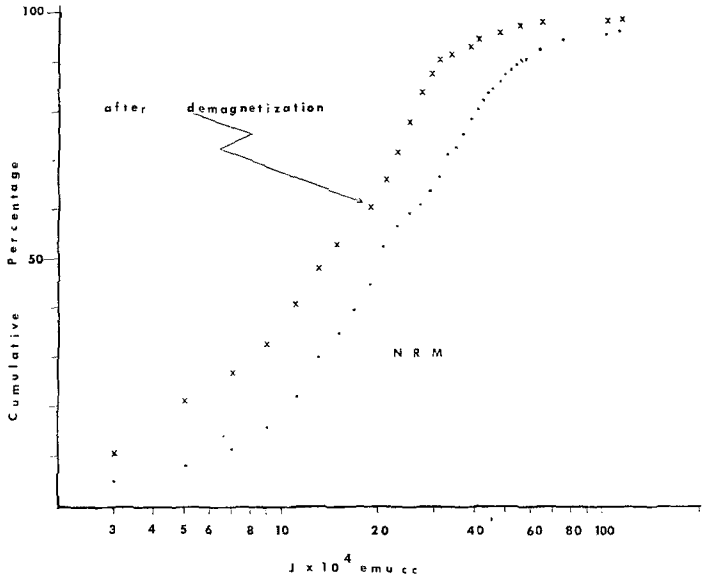


Fig. 6. Distribution of the itensity of magnetization in basalt samples from Lower Silesia shown as a cumulative percentage. The intensity scale is logarithmic. ● N. R. M. of all samples, x intensity after demagnetization.

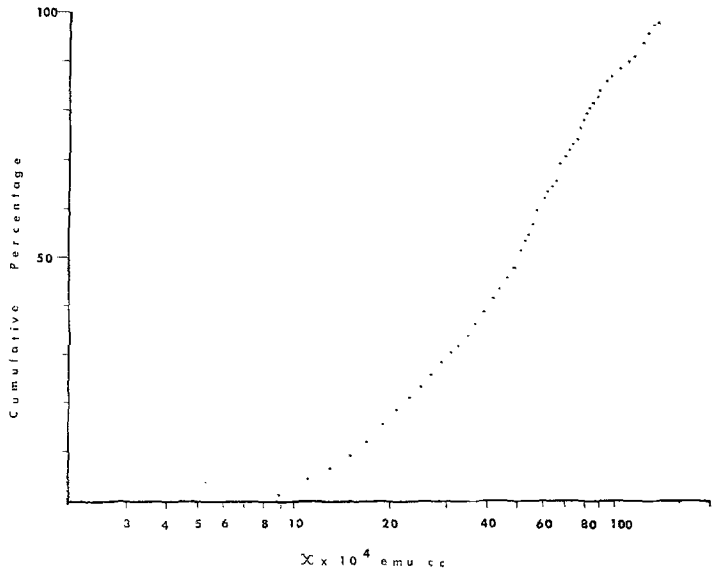


Fig. 7. Distribution of the magnetic susceptibility in basalt samples from Lower Silesia, shown as a cumulative percentage. The susceptibility scale is logarithmic.

(1964). This robs some of the value of single demagnetization curves from a site.

Both the intensity and susceptibility distributions show the now familiar close approximation to log normal curves. The absence of inflection points, if the few percent of high intensity high susceptibility samples are excluded, may suggest that the suite of samples was derived from a magma or magmas which behave in a uniform manner, at least in so far as the magnetic minerals are concerned.

Conclusions

The interpretation of the results of the Polish basalts from Lower Silesia is handicapped by the lack of stratigraphical control and the almost total absence of radiometric dates. Both the Silesian and the German Lausitz volcanics are part of the northern Bohemian province. It is worth remarking that the onset of vulcanicity in both regions seems to have begun in Upper Oligocene, and by inference this may be dated as older than upper middle Chattian. The youngest rocks at Lądek Zdrój are regarded as upper Pliocene or Pleistocene, and the positive magnetization would favour the latter. The location of this young activity suggests that there is no pattern of time migration of the centre of volcanic activity.

Of the majority of sites examined which appear to be Miocene, no further conclusions are possible at this stage.

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